

The Definitive Guide to the ARM Cortex-M3

Second Edition

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Joseph Yiu



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Foreword

Progress in the ARM microcontroller community since the publication of the first edition of this book has been impressive, significantly exceeding our expectations and it is no exaggeration to say that it is revolutionizing the world of Microcontroller Units (MCUs). There are many thousands of end users of ARM-powered MCUs, making it the fastest growing MCU technology on the market. As such, the second edition of Joseph's book is very timely and provides a good opportunity to present updated information on MCU technology.

As a community, progress has been made in many important areas including the number of companies building Cortex™-M3 processor-based devices (now over 30), development of the Cortex Microcontroller Software Interface Standard (CMSIS) enabling simpler code portability between Cortex processors and silicon vendors, improved versions of development tool chains, and the release of the Cortex-M0 processor to take ARM MCUs into even the lowest cost designs.

With such a rate of change it is certainly an exciting time to be developing embedded solutions based on the Cortex-M3 processor!

—**Richard York**

Director of Product Marketing, ARM

Foreword

Microcontroller programmers, by nature, are truly resourceful beings. From a fixed design, they create fantastic new products by using the microcontroller in a unique way. Constantly, they demand highly efficient computing from the most frugal of system designs. The primary ingredient used to perform this alchemy is the tool chain environment, and it is for this reason that engineers from ARM's own tool chain division joined forces with CPU designers to form a team that would rationalize, simplify, and improve the ARM7TDMI processor design.

The result of this combination, the ARM Cortex™-M3, represents an exciting development to the original ARM architecture. The device blends the best features from the 32-bit ARM architecture with the highly successful Thumb-2 instruction set design while adding several new capabilities. Despite these changes, the Cortex-M3 retains a simplified programmer's model that will be easily recognizable to all existing ARM aficionados.

—**Wayne Lyons**

Director of Embedded Solutions, ARM

Preface

This book is for both hardware and software engineers who are interested in the ARM Cortex™-M3 processor. The *Cortex-M3 Technical Reference Manual* (TRM) and the *ARMv7-M Architecture Application Level Reference Manual* already provide lots of information on this processor, but they are very detailed and can be challenging for novice readers.

This book is intended to be a lighter read for programmers, embedded product designers, system-on-chip (SoC) engineers, electronics enthusiasts, academic researchers, and others who are investigating the Cortex-M3 processor, with some experience of microcontrollers or microprocessors. The text includes an introduction to the architecture, an instruction set summary, examples of some instructions, information on hardware features, and an overview of the processor's advanced debug system. It also provides application examples, including basic steps in software development for the Cortex-M3 processor using ARM tools as well as the Gnu's Not Unix tool chain. This book is also suitable for engineers who are migrating their software from ARM7TDMI to the Cortex-M3 processor because it covers the differences between the two processors, and the porting of application software from the ARM7TDMI to the Cortex-M3.

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Finally, a special thank-you to Peter Cole and Ivan Yardley for their continuous support and advice during this project.

Conventions

Various typographical conventions have been used in this book, as follows:

- Normal assembly program codes:
`MOV R0, R1; Move data from Register R1 to Register R0`
- Assembly code in generalized syntax; items inside `<>` must be replaced by real register names:
`MRS <reg>, <special_reg>`
- C program codes:
`for (i=0;i<3;i++) { func1(); }`
- Pseudocode:
`if (a > b) { ...`
- Values:
 1. `4'hC`, `0x123` are both hexadecimal values
 2. `#3` indicates item number 3 (e.g., `IRQ #3` means `IRQ` number 3)
 3. `#immed_12` refers to 12-bit immediate data
- Register bits:
Typically used to illustrate a part of a value based on bit position; for example, `bit[15:12]` means bit number 15 down to 12.
- Register access types are as follows:
 1. R is Read only
 2. W is Write only
 3. R/W is Read or Write accessible
 4. R/Wc is Readable and clear by a Write access

Terms and Abbreviations

Abbreviation	Meaning
ADK	AMBA Design Kit
AHB	Advanced High-Performance Bus
AHB-AP	AHB Access Port
AMBA	Advanced Microcontroller Bus Architecture
APB	Advanced Peripheral Bus
ARM ARM	ARM Architecture Reference Manual
ASIC	Application-specific integrated circuit
ATB	Advanced Trace Bus
BE8	Byte-invariant big endian mode
CMSIS	Cortex Microcontroller Software Interface Standard
CPI	Cycles per instruction
CPU	Central processing unit
CS3	CodeSourcery Common Start-up Code Sequence
DAP	Debug Access Port
DSP	Digital Signal Processor/Digital Signal Processing
DWT	Data Watchpoint and Trace unit
EABI/ABI	Embedded application binary interface
ETM	Embedded Trace Macrocell
FPB	Flash Patch and Breakpoint unit
FPGA	Field Programmable Gate Array
FSR	Fault status register
HTM	CoreSight AHB Trace Macrocell
ICE	In-circuit emulator
IDE	Integrated Development Environment
IRQ	Interrupt Request (normally refers to external interrupts)
ISA	Instruction set architecture
ISR	Interrupt Service Routine
ITM	Instrumentation Trace Macrocell
JTAG	Joint Test Action Group (a standard of test/debug interfaces)
JTAG-DP	JTAG Debug Port
LR	Link register
LSB	Least Significant Bit
LSU	Load/store unit
MCU	Microcontroller Unit
MDK-ARM	Keil Microcontroller Development Kit for ARM
MMU	Memory management unit
MPU	Memory Protection Unit
MSB	Most Significant Bit
MSP	Main Stack Pointer
NMI	Nonmaskable interrupt

NVIC	Nested Vectored Interrupt Controller
OS	Operating system
PC	Program counter
PMU	Power management unit
PSP	Process Stack Pointer
PPB	Private Peripheral Bus
PSR	Program Status Register
SCB	System control block
SCS	System control space
SIMD	Single Instruction, Multiple Data
SoC	System-on-Chip
SP	Stack pointer
SRPG	State retention power gating
SW	Serial-Wire
SW-DP	Serial-Wire Debug Port
SWJ-DP	Serial-Wire JTAG Debug Port
SWV	Serial-Wire Viewer (an operation mode of TPIU)
TCM	Tightly coupled memory (Cortex-M1 feature)
TPA	Trace Port Analyzer
TPIU	Trace Port Interface Unit
TRM	Technical Reference Manual
UAL	Unified Assembly Language
UART	Universal Asynchronous Receiver Transmitter
WIC	Wakeup Interrupt Controller